The role of Camera-Based rPPG in Musical Biofeedback Systems for Stress Reduction and Consciousness Regulation

Fabian Winkelmann

Abstract—Hier kommt der Abstract hin.

Index Terms—Remote Photoplethysmography, Computer Vision, Musical Biofeedback, Stress Reduction, Heart Rate Variability, Self-Regulation, Consciousness States

1 Introduction

Stress can have detrimental effects on the human body, ranging from depression, anxiety and social withdrawal to sleep deprivation and obesity. Stress has a direct impact on the cardiovascular system [1]. A wide variety of methods exist to monitor these cardiovascular signals and to track stress levels. The measurement of pulse waves is called plethysmography. Photoplethysmography (PPG) utilizes light reflectance to detect physiological changes. A traditional method for PPG is contact-based sensors, such as pulse oximeters [2]. Remote Photoplethysmography (rPPG) uses standard video cameras to monitor the skin color and filter for subtle skin-color variations that are caused by flowing blood. Based on this information, the heart rate can be calculated [2]. These measurements can be directly reported back to the user using biofeedback. Biofeedback describes a therapeutic concept in which a user receives real-time information about their physiological processes, such as respiration or heart rate. This data is typically made accessible through an output device, such as a visual display. Alternatively, the information can be translated into different forms, such as auditory signals. The primary goal of biofeedback is to make these biological processes perceptible, enabling the user to gain conscious awareness and control over functions that were previously automatic [3].

The objective of this paper is to evaluate such a camera-based stress tracking system combined with musical biofeedback. The goal is to analyze if such a system can be used to lower stress levels and regulate consciousness. In the second chapter theoretical foundations such as stress, consciousness and rPPG techniques are investigated. In the third chapter, case studies will be analyzed. The fourth chapter evaluates whether such a system is theoretically feasible.

2 Stress and the autonomic nervous system

This chapter starts with a deeper look at stress and its impact on the autonomic nervous system. It explains why stress has an impact on the cardiovascular system. Moreover, it differentiates between the heart rate and the heart rate variability.

The autonomic nervous system consists of the parasympathetic and the sympathetic branches. The parasympathetic system acts during periods of rest and calm and regulates functions such as rest and digest. In contrast, the sympathetic system is activated in situations in which we perceive a threat. In these cases, the heart rate increases to release more energy [4].

McEwen (2007) describes this as the allostatic effect. The body dynamically adjusts its internal state to meet the standards of the environment. For instance, in a life threatening situation a lot of energy is released. The short-term mobilization of energy is a protective reaction of the body. If this activation occurs chronically without enough recovery, it can lead to permenant damage (Figure 1) [1].

Individual differences in stress perception are rooted in the brain's ability to evaluate environmental stimuli. According to McEwen (2007), the brain exhibits significant plasticity, meaning its responses to stressors are shaped by genetics, early development, and past experiences. If an individual has for example experienced a trauma, the brain's system may become hypersensitive [1]. It might trigger a 'fight or flight' response in harmless situations [4]. This can lead to coping strategies, such as smoking or drinking. These coping strategies try to calm down the sympathetic branch of the autonomic nervous system. On the other hand, the plasticity also allows for an intervention in the other direction. The brain can be trained to shift from a threatening state to a calm state [1].

When monitoring the heart, it is crucial to differentiate between the heart rate and the heart rate variability. The heart rate can be thought of as the quantity and the heart rate variability as the quality of heart activity [5].

The heart rate measures the total number of heartbeats per minute. It reflects how fast the heart is beating at any given moment. However, it can be described as a relatively coarse metric. Therefore, it is a releative insensitive marker for early-stage stress [5].

Heart Rate Variability (HRV), on the other hand, focuses on the smaller level of cardiac timing. It describes the

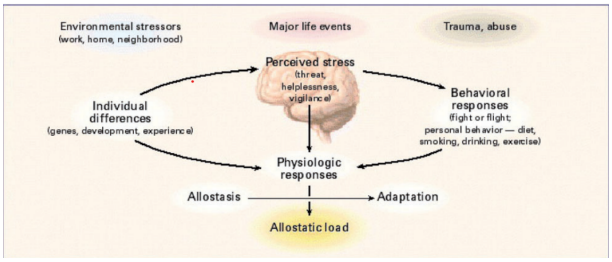

Fig. 1. The figure illustrates the mechanism of allostatic load and the underlying pathways of stress perception and physiological adaptation [1].

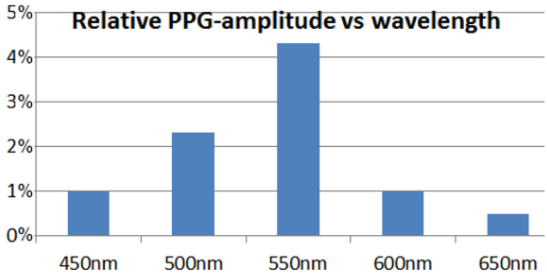

Fig. 2. Spectral sensitivity of the PPG signal. Green light exhibits the highest amplitude, making it the most reliable channel for non-contact heart rate monitoring. [6]

fluctuation in the time intervals between two heartbeats. This is defined as Interbeat Intervals (IBIs). HRV consists of the constant changes in these time intervals from one beat to the next [5].

From a physiological standpoint, a healthy biological system needs to be highly adaptable. This means that a healthy heart does not maintain a perfectly rigid or identical time between intervals. Instead, the autonomic nervous system constantly changes the timing of each beat to respond to internal and external demands [5].

If the intervals between heartbeats become too consistent and lose this natural irregularity, it is often a clinical sign of physiological strain or a lack of regulatory flexibility. High variability indicates that the body is in a state of balance and can rapidly adjust to stressors. Therefore, while HR measures the quantity of beats, HRV reflects the quality and efficiency of the autonomic nervous system's control over the cardiovascular system [5].

2.1 Remote Photoplethysmography (rPPG)

To measure the heart beat and the heart beat variability a technology called remote Photoplethysmography can be employed.

2.1.1 The foundations of rPPG

The foundational work for modern remote photoplethysmography (rPPG) was made by Verkruysse et al. (2008), who demonstrated that the heart beat could be monitored using a standard digital camera. This relies on the fact that hemoglobin absords light differently than the surrounding body tissue. As the blood volume changes in the body with every heartbeat, the intensity of the reflected light varies. Their research identified that the green color channel contains the strongest information regarding the heart beat (Figure 2). Despite the successful proof-of-concept, two challenges were highlighted: Motion artifcats caused by movement and noise due to the camera's sensor [2].

2.1.2 Evolution: From Classic Algorithms to AI-based Detection

Following the initial proof-of-concept, a significant increase in rPPG research was observed, particularly after 2017. This trend is closely linked to the rapid improvements in Artificial Intelligence (AI) and Deep Learning. A recent meta-analysis revealed that in 2025 82.05% [7] of papers about rPPG technology used an approach with artificial intelligence. A major problem is the availability of diverse, large-scale datasets. The available sets only have data of people from North America, Europe and China. This leads to a huge underrepresentation of various age groups and diverse skin tones [7].

2.1.3 Tradtional Frameworks and Region of Interest (ROI)

Traditional rPPG frameworks typically follow a multi-stage process. First, the face is detected and tracked, often using the Viola-Jones algorithm or similar cascade classifiers to define a Region of Interest (ROI). Then methods such as Principal component Analysis (PCA) or Independet Component Analysis (ICA) or employed to isolate the pulse signal from the green color channel. While effective if the person remains still, these approaches struggle with motion artifacts, as they require several seconds of stable data to compute the signal components [7].

While traditional frameworks rely on fixed mathematical models, modern approaches increasingly replace stages of the rPPG pipeline with Deep Learning modules. This is done in the area of face detection and tracking. The Viola-Jones algorithm often struggles with varying lighting conditions and extreme head poses. Current systems frequently utilize more robust AI-based detectors [7].

Furthermore, the region of interest (ROI) significantly impact the accuracy of the heart rate. Although, early studies often used the entire facial area, recent studies recommend using a small rectangular area at the forhead for better results [7].

2.1.4 ROI selection for remote photoplethysmography

The selection of the right Region of Interest (ROI) for rPPG measurements is critical for the suppression of motion and illumination artifacts. Previous approaches often utilized the entire face as a global ROI for signal extraction. However, Lempe et al. (2013) demonstrated that the signal quality varies substantially across different facial areas [8].

To quantify this, they implemented a grid layout over the face and calculated the local Signal-to-Noise Ratio (SNR) for each segment. Through this spatial analysis, they identified the cheek and forehead regions as the most promising areas for pulse extraction (Figure 3) [8].

While the forehead provides a stable signal, it has the practical drawback of often being covered by hair (e.g., bangs). These findings highlight a significant technical challenge. The requirement for advanced segmentation and

tracking algorithms that can reliably detect and follow the cheek area, even during head movements, to ensure a consistent and high-quality pulse signal [8].

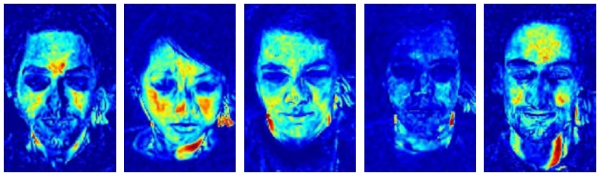

Fig. 3. The quality of the rPPG signal in the face. Blue shows low activity and red hight activity [8]

2.1.5 direkte Landmarks

A central problem of rPPG tracking is noise due to motion and light changes. Especially in the use case of human-computer interaction, brightness changes lead to severe fluctuations of pixel values in the face; examples include watching a movie or gaming. Li et al. (2014) propose a framework to solve these issues. First of all, the Viola-Jones detector is used to get a rough estimation of the face location. Then, 66 facial landmarks are placed, and based on these landmarks, a precise mask for the region of interest (ROI) is generated. KLT (Kanade-Lucas-Tomasi) points are used to continuously track the face, ensuring it is also tracked when moved. A second important step is the correction of brightness changes. It is assumed that these changes are not only visible in the face but also in the background. The average brightness changes in both the background and the face are measured, and a normalized least mean squares (NLMS) filter is used to find the optimal coefficient to subtract background noise from the facial area. Furthermore, non-rigid movement like smiling or speaking induces changes in the skin which cannot be compensated through tracking alone. Consequently, the measurements are split into small segments, and those with a high standard deviation are discarded. The remaining signal is processed with a bandpass filter to only allow for frequencies of 0.7 to 4 Hz (42–240 BPM). This framework was tested with the sophisticated MAHNOB-HCI database and reached a low error score of 6.87% with 527 users. Despite these achievements, there are still quite a few problems; for instance, large head movements lead to a significant loss of KLT tracker points, which results in an unstable localization of the ROI [9].

2.2 mediapipe

There are different approaches that use artificial intelligence to detect the face and the region of interest. Previous approaches used only 68 vertices in the face. A new approach, presented by Kartynnik et al. (2019), uses 468 vertices and is highly efficient. This model creates a 3D mesh of the face, meaning it can estimate the depth of each point even from a normal 2D camera 4. Depending on the device, it can deliver 100 to 1000+ FPS [10].

Different models that also use many vertices often use standard methods based on averaging effects. They don't work well when there are small inaccuracies, like the closing of one eye. Due to the architecture of the neural network developed by Kartynnik and his team, it is possible to use it in real-time. It has been trained on a dataset of around 30,000 images. It has been tested that it has the same accuracy as human experts that model the face [10].

Fig. 4. Example of a face mesh calculating with an artificial intelligence model [10]

2.3 Blind Source Separation

The fundamental problem in rPPG is that the recorded RGB channels do not only contain the pulse information but also significant noise in the form of light changes and movement. The challenge is that this noise has to be separated from the real physiological information. Mathematically speaking, the resulting color channels are treated as a linear mixture [11] :

$$x(t) = As(t)$$

The goal is to find a demixing matrix W, so that the original source signals $\hat{s}(t) = Wx(t)$ can be calculated. To find W, it is important to search for statistical independence between the sources. For that, the JADE algorithm (Joint Approximate Diagonalization of Eigenmatrices) is used. Unlike deterministic methods, JADE is an iterative process that tries out different combinations to maximize non-Gaussianity, which results in a higher computational load and increased latency (Figure 5) [11].

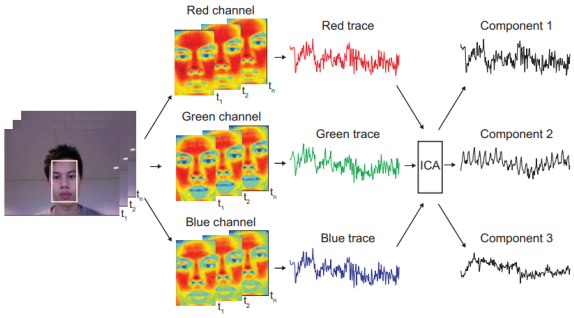

Fig. 5. Seperation of color channels [11]

For the initial localization, the Viola-Jones algorithm is used to detect the face. In this specific setup, only 60% of the bounding box width is used as the Region of Interest (ROI). The raw signals from this ROI are then spatially averaged and normalized using Zero-Mean and Unit-Variance to balance different channel intensities [11].

After the independent components are recovered, the pulse is extracted by using the Fast Fourier Transform (FFT). An operational range from [0.75 - 4.0] Hz is allowed, because this corresponds to 45 - 240 BPM, the range a human pulse usually lies in. Additionally, a temporal consistency threshold is implemented: a jump of only 12 BPM is allowed between consecutive measurements (taken 1 s apart). Anything different is rejected as noise or an artifact. The highest power peak within the frequency spectrum is chosen as the final heart rate [11].

To compare the accuracy of this methodology to a gold standard (like a finger BVP sensor), Bland-Altman plots are used. These plots show that ICA can significantly reduce the Root Mean Squared Error (RMSE) from 19 BPM to about 4.6 BPM even when the subject is moving [11].

However, there are physical limitations to consider. According to the Beer-Lambert Law, the way light travels through and is absorbed by blood vessels is technically non-linear. But for the sake of computation, ICA uses a linear approximation, which works well for short time windows (e.g., 30s). The system is also highly dependent on the ambient lumination; if it is too dark, the SNR drops too low for a reliable measurement [11].

Another risk is the "Component Picking" problem. In this study, the second independent component is usually chosen as the pulse signal because hemoglobin absorptivity is highest in the green spectrum. However, this fixed selection could be wrong if the noise is too strong. Finally, it is noted that using simple webcams is much more cost-effective than using expensive thermal cameras, and the software-based approach even allows for the simultaneous detection of multiple people in one frame [11].

2.3.1 Chrom approach

The Chrominance-based method (CHROM) introduced a physicsbased alternative. It solves the big challenge of subject motion, when calculating the heart beat. Periodic movements can render simple intensity-based selections useless. Furthermore, while many algorithms require long observation intervals to achieve sufficient frequency resolution, CHROM is designed to be highly efficient. Similarly to traditional methods like PCA or ICA, the chrom algorithm also uses the fact that the blood absorb a large percentage of green light. Additionally it analyzes how the light is reflected on the human skin. According to the dichromatic reflection model this reflection has a Specular and a Diffused Component (Figure 6). Only the diffuse Reflexion enters the skin and has the color of the blood. The specular reflexion does not enter the skin. It is just reflected on the surface and thus does not contain any information about the blood. This specular part is an additive value and thus the division of the two color channels does not cancel this effect out. Thereforce, a subtraktive approach is used [6].

$$(R + noise) - (G + noise) = R - G$$

The CHROM algorithm projects the normalized RGB signals into two orthogonal chrominance signals, X and Y. In a perfect environment under white light, these signals are mathematically independent. Based on empirical data from a diverse set of subjects, de Haan and Jeanne derived

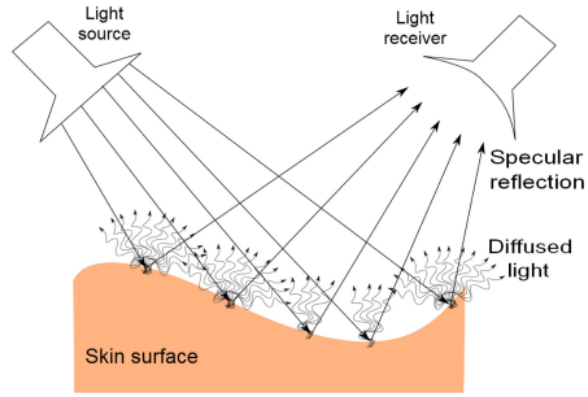

Fig. 6. The illustration depicts light reflection on the skin [6]

the standardized projection vectors $[0.7682, 0.5121, 0.3841]$, leading to the core transformations:

$$X_s = 3R_n - 2G_n$$

$$Y_s = 1.5R_n + G_n - 1.5B_n$$

To further refine the pulse signal S, these components are combined using the ratio of their standard deviations (α):

$$S = X_s - \alpha \cdot Y_s \quad where \quad \alpha = \frac{\sigma(X_s)}{\sigma(Y_s)}$$

Even after projection, the signal S may contain a slow "drift" caused by respiration or minor posture changes. To isolate the cardiac signal, a Bandpass Filter is applied. The filter is typically tuned to a range of 0.7 Hz to 4.0 Hz, which corresponds to a human heart rate between 42 and 240 BPM. Frequencies outside this window are discarded as non-physiological noise.To extract the final heart rate, a Fast Fourier Transform (FFT) is performed on the filtered signal. The FFT shifts the data from the time domain to the frequency domain, resulting in a power spectrum. The algorithm identifies the "dominant peak" (the highest frequency component) in this spectrum. For instance, a peak at 1.2Hz is converted to the final pulse rate:

$$1.2Hz \times 60seconds = 72BPM$$

One of the most significant advantages of CHROM is its temporal efficiency. While traditional BSS methods (Blind Source Separation) like ICA or PCA often require up to 30 seconds of data to stabilize, CHROM utilizes an Overlap-Add (WOLA) procedure. By using overlapping windows of only 1.6 seconds, the system provides a continuous, near-instantaneous pulse estimate.

In comparative studies, the robustness of CHROM is evident. CHROM has a high stationary accuracy and predictes the correct pulse with a 98% accuracy. It has a 92% correlation with the medical-grade contact sensor. Furthermore, it is resilient against motion. Under havey motion (e.g., on a fitness stepper), maintains a 48% success rate. ICA and PCA fail significantly, and only reach 4% and 11%. Moreover,

CHROM can also process different skin tones. Darker Skin Tones naturally lower the Signal-to-Noise Rratio because the absord more light. Chrom remains the most robust classical algorithm across all phenotypes.

2.3.2 Modern Deep Learning Paradigms

With the rapid advancement in artificial intelligence, modern rPPG systems have shifted towards end-to-end models. These architectures can process raw data or image sequences to estimate heart rate directly, often without the need for manual face isolation or region of intrest selection. There are three architectural paradigms: Convolutional Neural Networks (CNNs), Hybrid Neural Networks and Transformed-based architectures [7].

In modern rPPG systems, Convolutional Neural Networks (CNNs) are the standard for processing image data. These networks consist of several layers, mainly an input, an output, and various hidden layers like convolutional and pooling layers. Their main task is to recognize the very subtle color changes in the skin pixels that happen during a cardiac cycle [7].

A big problem in this field, is that there aren't enough large and diverse datasets to train these models from scratch. Because of this, researchers often use Transfer Learning [7]. The idea here is to take a model that was already trained on a different task, such as general face recognition. By 'freezing' the first layers and only fine-tuning the last ones with rPPG-specific data, the model can use its existing knowledge of facial features. This approach makes it possible to get good results even with smaller datasets and saves a lot of training time [12].

In addition, a different category of neural networks has emerged: Hybrid Neural Networks. These combine different neural networks like CNNs, RNNs and autoencoders to form a more robust system [7].

In the past years powerful Large Language Models (LLMS) like ChatGPT have emerged that use transformers to understand words and sentences. The same technique can be employed to analyze pictures. The picture is broken up into small rectangular patches, which the network analyzes in relation to one another [13]. By Comparing different regions simultaneously, the transformer is able to differentiate between the heart beat and noise, such as micro-movements. This leads to a more robust heart rate estimation [7].

2.3.3 Summary of rPPG Frameworks

There are lot of different approaches to calculate the heartbeat with an rPPG technique. Modern Deep Learning Algorithms have emerged in the recent years. They show promising results, although there are not enough diverse datasets. The CHROM Algorithms is promising for a real-time evaluation of data. Traditionally algorithms like PCA are slow and not so precise. Although, there has been a huge advancement in the past years, there are still no long-term studies that monitor the heart beat over several hours. Most researches only test rPPG for a short time, which does not show how it works in real daily life [7].

2.4 sonification

Dubus et al. (2013) performed a systematic review on sonification. They mention that sonification can enhance the perception of one's own body and is therefore widely used in rehabilitation and training. Among different sonification techniques, parameter mapping is especially promising because it allows for continuous transmission of information. In this process, a certain set of mappings between "physical dimensions" and "auditory dimensions" is defined [14].

To establish these mappings, a three-stage process is required. First, the specific auditory dimensions to be used must be identified, such as pitch or timbre. Then the polarity has to be defined. This determines the direction in which the auditory dimension changes according to the data. For instance, if the heart rate increases, it must be decided whether the pitch should increase or decrease. Finally, the psychophysical scaling has to be addressed. Since humans perceive changes in sound non-linearly, it is necessary to define the magnitude of the auditory change relative to the data change [14] [15].

According to Dubus and Bresin (2013), many sonification approaches implement a strategy called "ecological perception". This means that certain sonification principles are part of our subconscious mind. This applies, for example, to loudness, which can be explained by the inverse distance law: a decrease in sound intensity is naturally associated with an increasing distance from a source. Such natural associations are deeply ingrained in human perception and should be prioritized when employing a sonification approach [14].

Research has shown which auditory dimensions are preferred in specific fields. For the dimension "Time", timbre is often used, whereas loudness is used less frequently. In the category of "Kinetics", which describes energy or force, loudness is the most common mapping. These categories are particularly relevant when sonifying the human heartbeat. Furthermore, in real-time sonification, a mapping from "instant to instant" occurs, meaning the rhythm of the underlying data source is directly incorporated into the auditory output. Additionally, data related to horizontality is more often depicted through spatialization, while vertical data is frequently sonified through pitch [14].

Furthermore, it is pointed out that the sonic material is decisive. If the sounds make the user uncomfortable, this can have drastic consequences for the effectiveness of the system. Therefore, the use of sample-based displays is often preferred over simple synthetic tones [14].

In addition, it is important to investigate the chosen sonification strategy scientifically. The study highlights that many approaches are currently chosen in an ad hoc or arbitrary manner without verifying their effectiveness through objective tests [14].

2.5 Mapping of heart beat to music

RPPG Techniques provide the functionality to measure the heart beat remotely. These measurements then need to be converted to music.

There has been some research on passive music listening without any biofeedback mechanism. According to a meta-analysis by Pelletier(2004), music consistently assists in lowering stress levels. The magnitude of this response is affected by several factors. Patients with prior musical experience, such as playing an instrument, tend to lower

their stress levels a lot more. Furthermore, Pelletier notes that indivudal therapy generally results in better relaxation than group session. Although, group sessions are used much more often because they are less expensive. A huge obstacle is that the devices that are used for data collection often cause discomfort. This can lead to false data regarding the relaxation process. Finally, the study suggests that music based on scientific research (e.g., specific rhythmic or frequency patterns) is more effective for stress reduction that music chosen soley based on personal preference [16]. In another study, Linnemann et al. examined how music affects stress in daily life rather than in a laboratory. They came to the conclusion that listening to music can effectively reduce stress levels. Nevertheless, the largest effect is reached when subjects listen to music with then intention of decreasing stress levels. This suggests that the reason why someons listens to music is very important for the actual stress-reducing effect [17].

These results are also supported by another study [18]. They confirm that listening to music alone does not automatically decrease stress levels. Instead, the reason of listening to the music is the most important factor. For example, if someone listens to music to cope with loneliness, it might effectively decrease their feeling of isolation. Nevertheless, if the goal is relaxation, the music only works well if the listener has the specific intention to relax [18].

The study of Bernardi et al. (2006) confirms that the results do not depend on the music preferation of the user rather on the BPM and rhythm structure of the music. Additionally, the study found no evidence of habituation, meaning the users do not simply 'get used' to the music over time. In Contrast to Linnemann et al. (2015), who emphasized the importance of the listener's intention, Bernadi concludes that music can be perceived as a passive form of meditaiton. consequently, this means that the user does not need an intend to reduce stress for measurable results [19].

Passive music listening without any biofeedback system has mixed results. To convert the heartbeat to music, a process called mapping is necessary, that takes the input data and transforms it into music. There are two methods Continous mapping and diskrete mapping. Continous mapping describes the process of constantly changing the music according to the data. During the discrete mapping process certain aspects of the music are changed when a certain threshold is reached [20].

2.5.1 Biofeedback-Driven Regeneration

As explained previously the heart rate is converted to a piece of music. It is now crucial to understand how the music can help the human body to regenerate. According to Lehrer and Gevirtz (2014), the efficiency of biofeedback relies on the synchronization of the cardiovascular and respiration systems [21].

The human cardiovascular system has a certain resonance frequency at which point it exhibts the best effects on the human body. If this state is reached, has maximum internal order and physiological efficiency. This frequency is reached at approximately 0.1 Hz. This is equivalent to roughly six breaths per minute. By breathing at the resonance frequency, heart rate and blood pressure enter a harmonic state. Furthermore, gas exchange in the lungs are also optimised at this state [21].

In addition, the activity of the vagus nerve is stimulated. The vagues nerve is responsible for the communication between the heart and the blood. This makes it possible for the body to lower the heart rate more effectively [21].

Moreover, biofeedback creates a meditative effect that changes the user's state of consciousness. The user has to focus his thoughts and his energy on the biofeedback signal. Thus, it the brain cannot focus on anxious thoughts [21].

2.6 Consciousness and State Regulation

The meditative effect of focusing on one's heart rate or respiration can be used to enhance consciousness. This process is frequently described through the concept of mindfulness. This makes it possible for individuals to develop a sensitivity to their thoughts and emotional states. Bishop et al. (2004) established this definition. A fundamental requirements is described as the 'non-judgemental-approach", meaning. This means that internal states are perceived with curiosity and openness. Internal thoughts and feeling should be accepted as they are [22].

3 CASE STUDIES

In this chapter two case studies will be presented that highlight the effects of using biofeedback. The use different approaches. One uses the heart beat and the other uses EEG. They are both important for validating stress reduction and consciousness regulation.

3.1 The "Unwind" Musical Biofeedback System

The "Unwind" system, developed by Yu et al. (2018), employs a special approach to mapping the heart rate to music. It uses nature soundscapes combined with sedative music. Sedative music is used in music therapy to effectively decrease heart rate and blood pressure. It has a low tempo and a repetitive structure [23].

The system uses the heart rate variability to provide feedback on the user's breathing patterns. The heart rate accelerates during inhalation and decelerates during exhalation. The framework maps these fluctuations directly to a wind sounscape. The wind volume is reduced during inhalation and during exhalation the wind becomes louder. The user should adapt his breahting pattern in a way that a the wind sound is like a smooth, consistent sinve wave. This promotes consistent respiration [23].

The wind provides immediate feedback on the breathing pattern. The overall relaxation is mapped with a special form of heart rate variability. The system calculates the variability with a sliding window of 16 heartbeats. The organic soundscape is based on the intensity of the Heart Rate Variability. If the variability is low, the environment becomes busy and crowded with bird calls and noises. On the other hand, if the variability increases, the soundscape becomes simpler and more transparent. The frequency of animal sounds decreases and the overall volume drops. This rewards the person with acoustic clarity [23].

The system was validated with 40 young adults. A significant decrease in heart rate was found in the music

assisted conditions, and the reduction in respiration rate was significantly greater with biofeedback than without. Skin conductance responses, which indicate sympathetic nervous system activity, decreased markedly during the relaxation sessions. In follow-up interviews, 70% of participants reported that the nature sounds allowed them to quickly visualize calming natural scenes, aiding their focus and mood [23].

3.2 Mindfulness

This study investigated if a system with biofeedback could lower the entry barriers of classical meditation. The focus was on improvements in emotional regulation and cognitive performance. The experiment took place over four-weeks and involved 40 participants that suffered from mild to moderate stress. The participants were separated into an experimental group and an active control group. As a tracking device, this experiment utilized the Muse™ headband (Figure 7). This is a mobile EEG system connected to a smartphone app. The participants had to do daily, active attention traning sessions. The particiapnts were able to gather real-time feedback with the Muse™ headband. The recorded brainwaves were translated into acoustic feedback. The active control group performed identical relaxation exercises for the same duration but without the wearable device or acoustic feedback [24].

Fig. 7. Subject wearing the Muse™ EEG headband during a neurofeedback session. [24]

The results after four weeks were significant and unserscore the potential of wearables in professional stress management. The control group did not show large changes. On the other hand, the experimental group recorded a decrease in perceived stress of 7% after two weeks. After weeks it went down to 16%. In addition, participants demonstrated higher concentration levels and better impuls control during complex tasks. Moreover, the participants' brainwaves showed a lasting shift toward a relaxed mind. The central finding of Balconi et al. is that real-time feedback is the decisive factor for success.

4 ARGUMENTATION

In this chapter, the final argumentation for the closed-loop system is presented. It discusses whether a camera-based approach using rPPG combined with musical biofeedback is effective for stress reduction and the regulation of consciousness.

First and foremost, stress in this context is defined as the activation of the sympathetic nervous system and the corresponding withdrawal of the parasympathetic branch. These physiological shifts directly affect heart activity. While physical activity is best tracked via heart rate (HR), situations in which the user is sitting still or not moving significantly require the analysis of Heart Rate Variability (HRV) to accurately quantify stress. Consequently, a comprehensive system must be capable of addressing both states.

Both aspects can be effectively monitored using an rPPG approach. For real-time feedback, it is crucial that the underlying algorithms are computationally efficient. Furthermore, the system must remain robust against motion artifacts. In daily life, users do not remain perfectly still for extended periods; therefore, the CHROM algorithm is ideally suited for this use case. It is fast to calculate as it leverages the standardized optical properties of human skin, and it has been proven to be resilient against motion changes. In contrast, other methods such as PCA or ICA often require longer observation windows for an initial measurement or demand significant hardware resources. Deep Learning approaches frequently require a dedicated GPU and struggle with real-time processing, as they often need to analyze both past and future frames of a video sequence. Additionally, these networks are often trained on datasets that lack a sufficient variety of skin tones. This leads to the conclusion that a CHROM-based approach is best suited for this application.

In the next step, this physiological data must be converted into musical biofeedback. The music must be enjoyable to the user to prevent listening fatigue while also avoiding boredom. Passive listening alone is generally not successful in lowering stress; the user's intention is far more critical. The user must actively intend to lower their stress levels and gain better conscious awareness. A continuous mapping process is necessary to provide the user with immediate feedback regarding their own body.

The "Unwind" system successfully utilized nature sounds, such as wind noises, to imitate breathing patterns. Furthermore, an organic soundscape can be used to reflect the participant's arousal level. This follows the principle of "ecological perception" according to Dubus and Bresin (2013): if the body is calm, the soundscape reflects this tranquility.

Additionally, case studies involving the "Muse" headband have demonstrated that biofeedback can effectively increase conscious awareness and mitigate anxious thoughts. Biofeedback serves as an "anchor," preventing the user from ruminating on distracting or stressful thoughts. This process also enables various health-promoting mechanisms through biofeedback-driven regeneration. The respiratory and cardiovascular systems are synchronized, and when this resonance frequency is reached, the body functions at its optimum. This stimulates the vagus nerve, making it possible to further lower stress levels and promote long-term recovery.

5 CONCLUSION

Here goes the conclusion

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